22AIE304 DEEP LEARNING · FINAL REVIEW

Self-Supervised Masked Autoencoder Pretraining for Low-Label Automatic Modulation Classification Using IQ Signals

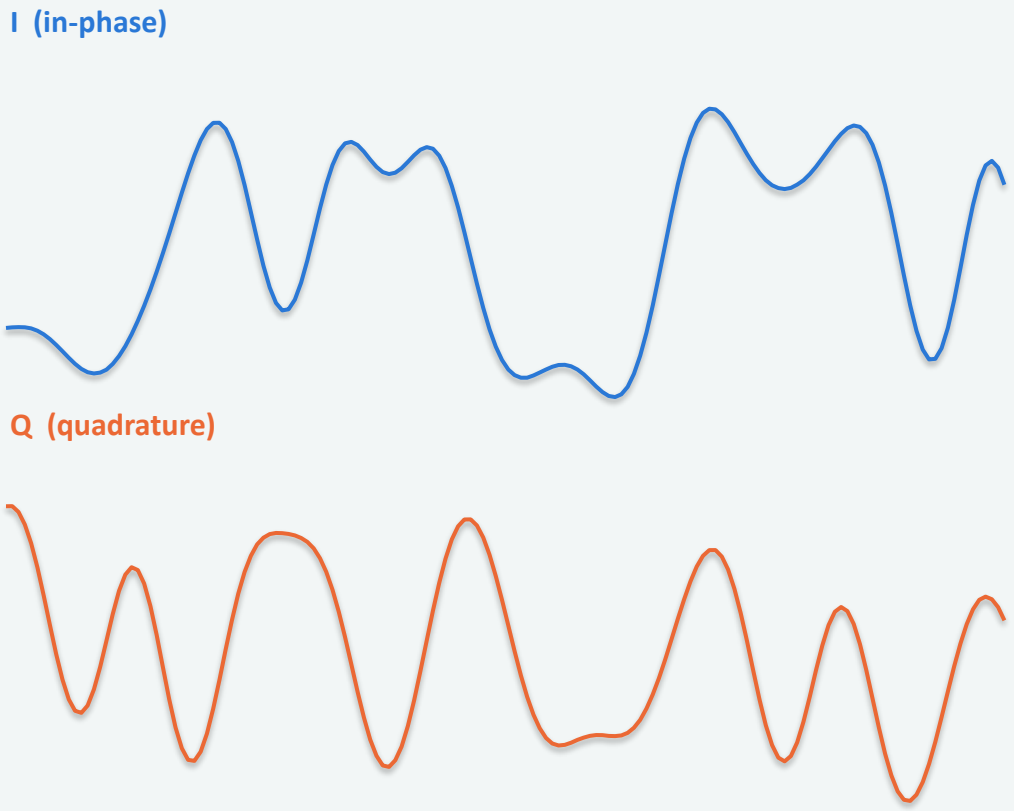
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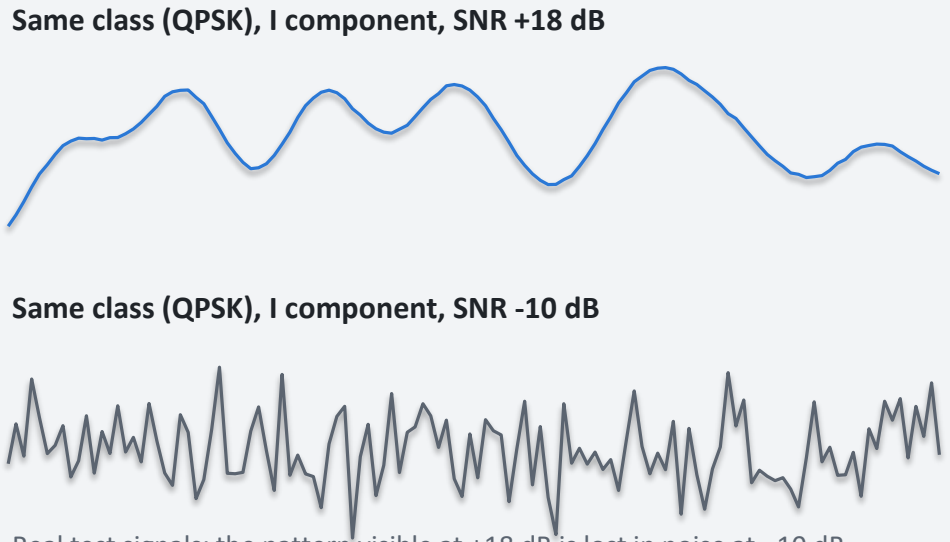
B.Tech CSE (Artificial Intelligence)



A real held-out test signal: QPSK at +30 dB SNR, first 160 of 1,024 samples (RadioML 2018.01A, test index 1,229)

Modulation is hard to recognise with few labels and low SNR

- **Automatic modulation classification (AMC):** identify how a received radio signal was modulated, e.g. QPSK or 16QAM, directly from its I/Q samples
- **Labels are expensive:** each training signal needs a known transmitter setting, RF equipment and expert effort
- **Low signal-to-noise ratio (SNR)** buries the modulation pattern in noise, so accuracy drops sharply
- **Unlabelled signals are plentiful.** Self-supervised pretraining can learn signal structure from them first, then fine-tune with few labels



Real test signals: the pattern visible at +18 dB is lost in noise at −10 dB

Research question: does self-supervised CNN masked-autoencoder (MAE) pretraining improve AMC when labelled data is limited, compared with training the same CNN from scratch?

Objectives: what we set out to do, and what was completed

1.	Build a CNN classifier trained from scratch (the baseline)	Completed
2.	Pretrain a CNN masked autoencoder on unlabelled raw IQ (1.79 M training signals)	Completed
3.	Compare both at 1%, 5%, 10%, 25% and 100% labelled data	Completed
4.	Analyse accuracy, macro-F1 and per-SNR behaviour	Completed
5.	Investigate longer classifier fine-tuning (50 and 100 epochs, 1% labels)	Completed · exploratory
6.	Cross-dataset transfer to RadioML 2016.10a	Completed · separate study
7.	Linear probe with a frozen encoder; test whether reconstruction error predicts accuracy	Not done

Objectives 6–7 came from the mid-review plan. The transfer study is reported separately (11 classes, 128-sample signals): the transferred encoder was 0.009–0.012 macro-F1 below scratch at every budget.

Raw-IQ MAE exists; a controlled evaluation is missing

Study	Approach	Relevance / limitation for our question
O'Shea, Roy & Clancy (2018)	Supervised deep CNN on raw IQ; introduced RadioML 2018.01A	Strong baseline, but needs fully labelled data
Liu et al. (2021)	Self-contrastive pretraining + fine-tuning	Uses unlabelled signals; contrastive, not masked reconstruction
He et al. (2022)	Masked autoencoders (ViT) for images	Established masked reconstruction as SSL; built for images
Shi et al. (2024), GAF-MAE	MAE on Gramian Angular Field images of signals	Signals converted to images, not raw IQ
Liu et al. (2025), RIS-MAE	MAE directly on raw IQ for AMC	Closest prior work: proposes the method; not our controlled multi-budget comparison

Gap we address: does raw-IQ MAE pretraining help **consistently** across label budgets, random seeds and SNR levels, when compared with a like-for-like scratch baseline that shares the same CNN backbone and training recipe?

NOVELTY AND CONTRIBUTIONS

Our contribution: a careful experiment, not a new architecture

Controlled comparison

Scratch vs CNN-MAE with the same backbone, data and fine-tuning recipe

Five label budgets

1%, 5%, 10%, 25%, 100%; nested, class-balanced subsets

Seed-level analysis

3 paired seeds at 1–10%; paired differences and a paired t-test

Per-SNR analysis

Accuracy across all 26 SNR levels, with the low-SNR band examined

Longer fine-tuning

Exploratory 50- and 100-epoch follow-ups at 1%, three seeds each

Reproducible evidence

Manifests and hashes, verified figures, a working Streamlit demo

Not claimed: a new MAE architecture, the first raw-IQ MAE (RIS-MAE, 2025, came first), or state-of-the-art accuracy. Our encoder is a small patch-based CNN (55,544 parameters).

DATASET

RadioML 2018.01A: 2.56 M labelled IQ signals, 24 classes

2,555,904
IQ signals

24
modulation classes

26
SNR levels, −20 to +30 dB (2 dB steps)

2 × 1,024
I/Q channels × time samples

Fixed split (70 / 15 / 15)



Label budget	1%	5%	10%	25%	100%
Labelled training signals	17,883	89,451	178,914	447,281	1,789,132
Seeds (primary)	3	3	3	1	1

Secondary: RadioML 2016.10a
220,000 signals · 11 classes · 2 × 128. Used only for the separate transfer study.

Source: DeepSig RadioML 2018.01A (O'Shea, Roy & Clancy, 2018; deepsig.ai/datasets). Validation for checkpoint selection uses a fixed 12,480-signal subset (20 per class × SNR) of the validation split.

Raw IQ input, fixed partitions, no test data in training



Leakage controls

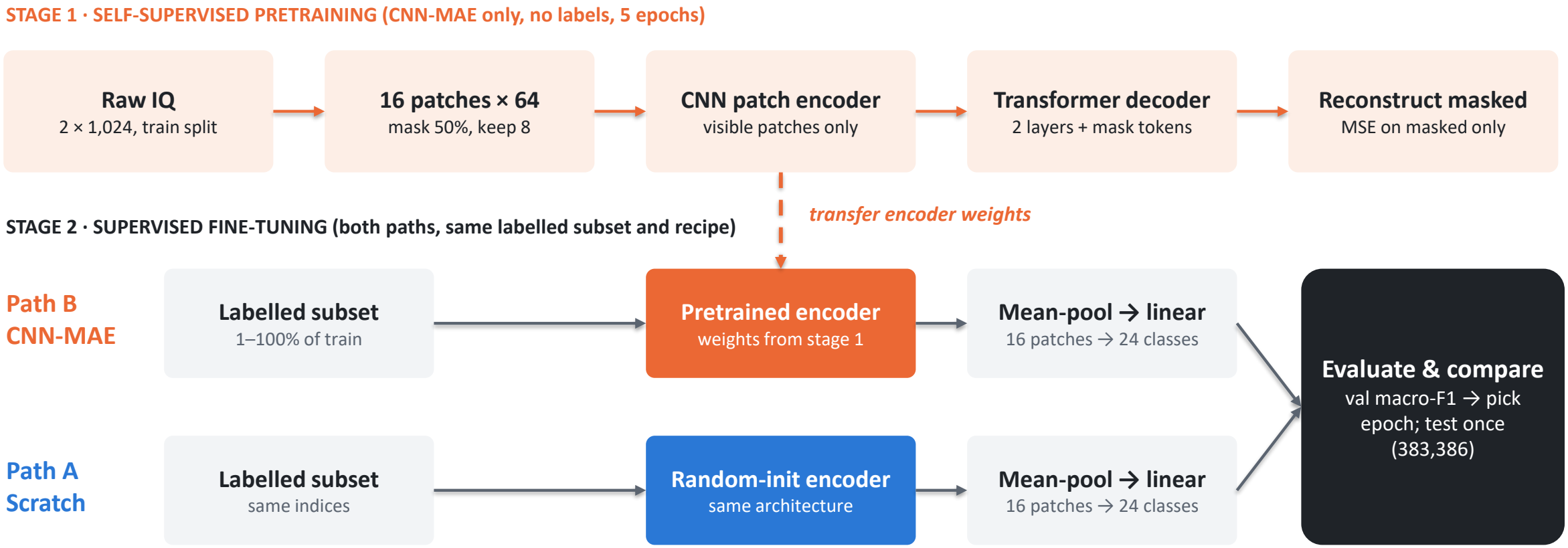
- Train, validation and test indices are disjoint (checked), and the same for every budget and seed
- MAE pretraining sees only training signals, without labels; its checkpoint is chosen by validation MSE
- Classifier checkpoints are chosen by validation macro-F1 on the fixed 12,480-signal subset
- The 383,386 test signals are evaluated once per selected model, after training

Checked vs not claimed

Loader rejects non-finite samples and invalid one-hot labels. Label subsets are balanced by class, not by SNR. No duplicate or missing-value filtering is performed.

PROPOSED SYSTEM ARCHITECTURE

One CNN backbone; the two paths differ only in encoder start



Encoder per 64-sample patch: 3 × (Conv1d → BatchNorm → GELU), max-pooling, global average pool, linear → 128-d. The whole classifier (encoder + head) is fine-tuned in both paths. The decoder and positional embeddings are used only in pretraining.

Same fine-tuning recipe; only the initialisation changes

- **Why a CNN:** 1-D convolutions learn local phase/amplitude patterns directly from raw I/Q
- **Self-supervised signal:** hide half of the 16 patches and reconstruct them from the visible half
- **Transfer:** copy the pretrained encoder; add a fresh linear head; fine-tune everything
- **Scratch baseline:** same model and recipe from random initialisation, so any gap is due to pretraining

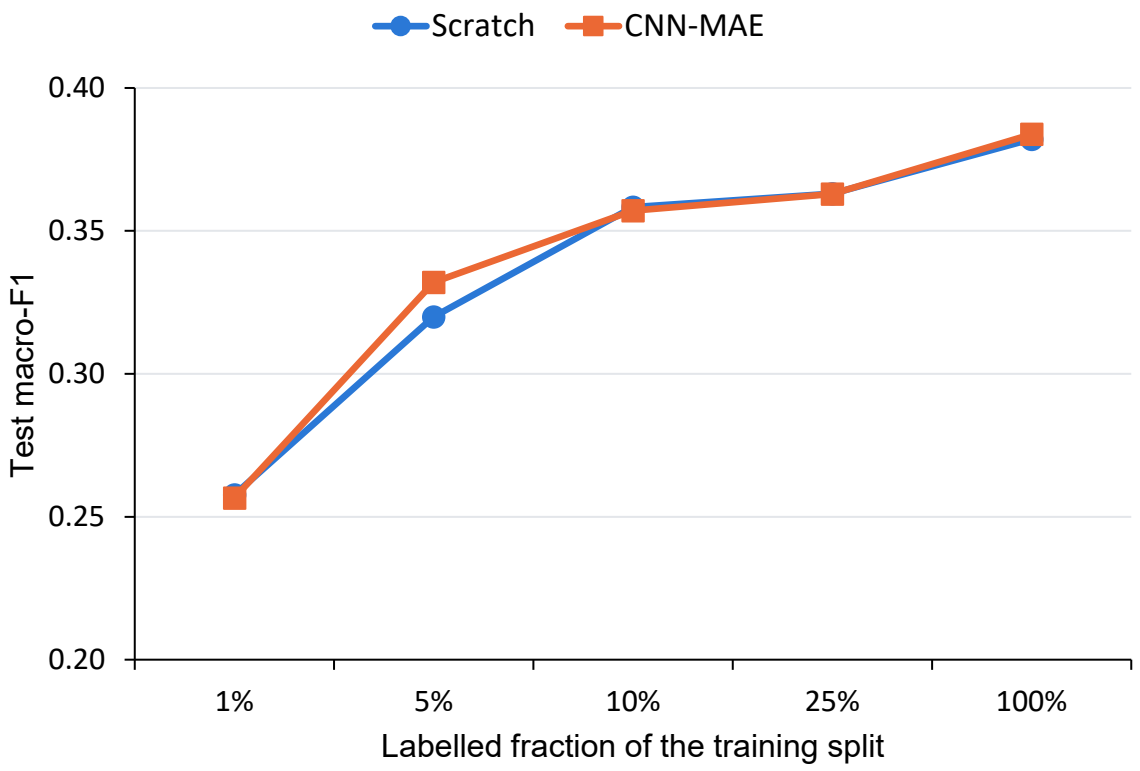
Setting	MAE pretraining	Classifier fine-tuning
Data	1,789,132 train signals, no labels	labelled subset (1–100%)
Loss	MSE on masked patches	cross-entropy
Optimiser	AdamW, lr 1e-4, wd 1e-4	AdamW, lr 1e-3, wd 1e-4
Schedule	cosine, 5 epochs, batch 64	cosine, batch 128, grad-clip 1.0, FP16
Epochs	5	**10 primary · 50 / 100 exploratory**
Selection	lowest validation MSE (epoch 5)	best validation macro-F1 (12,480)

Environment: Python 3.11, PyTorch 2.10.0 + CUDA 12.6, NVIDIA RTX 4050 Laptop GPU (6 GB). Pretraining took 4.9 h.

PRIMARY = 10 epochs, pre-specified

EXPLORATORY = 50 / 100 epochs, added later

Primary result: modest gain at 5%, no consistent advantage



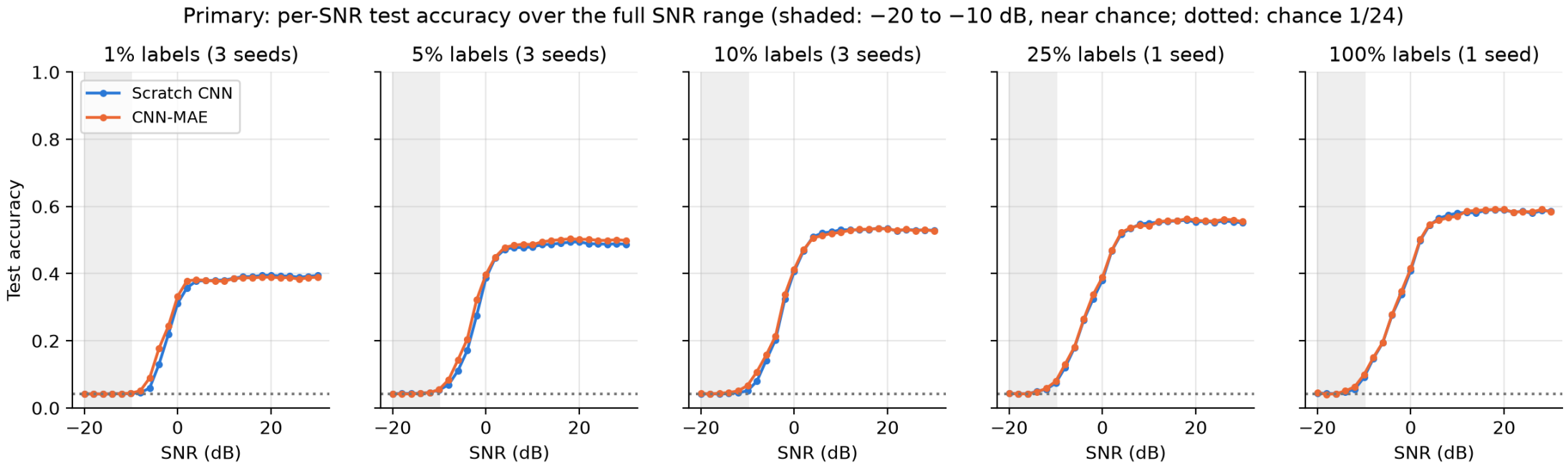
Mean over seeds (3 at 1–10%, 1 at 25% and 100%). The y-axis starts at 0.20.

Labels	Seeds	Scratch	CNN-MAE	Δ F1	Δ acc.
1%	3	0.2577	0.2566	-0.0011	+0.0041
5%	3	0.3199	0.3320	**+0.0121**	+0.0106
10%	3	0.3583	0.3571	-0.0012	+0.0031
25%	1	0.3630	0.3629	-0.0002	+0.0028
100%	1	0.3821	0.3838	+0.0017	+0.0017

Δ = CNN-MAE − scratch (F1 = macro-F1, acc. = accuracy). Macro-F1 = unweighted mean of per-class F1 over 24 classes.

5%: +0.0121 macro-F1 (2 of 3 seeds; paired t-test p = 0.29, not significant). At 1%, 10% and 25% MAE is marginally lower.

Pretraining helps most at −8 to 0 dB with few labels

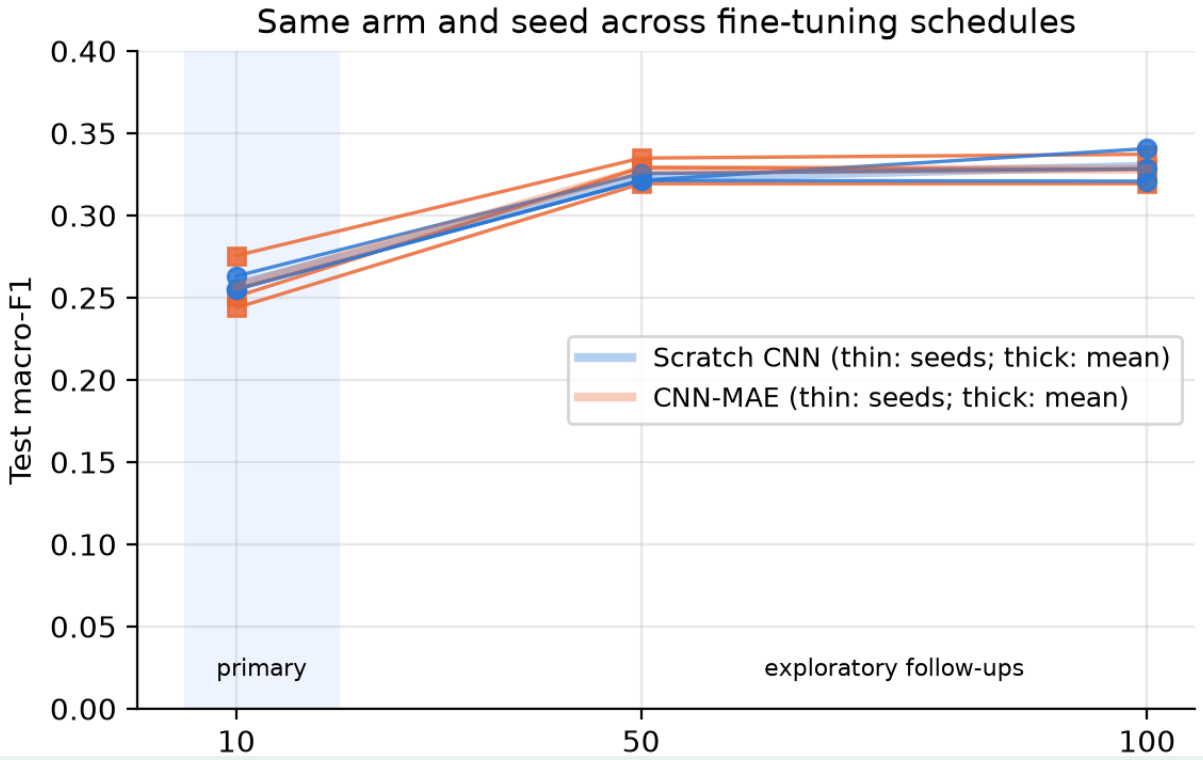


Labels	−20 to −10 dB acc. (scratch / MAE)	−8 to 0 dB: Δ acc.	≥ 10 dB: Δ acc.
1%	0.042 / 0.042	**+2.6 pp**	-0.4 pp
5%	0.045 / 0.045	**+2.8 pp**	+1.0 pp
10%	0.044 / 0.049	**+1.5 pp**	-0.1 pp
25%	0.051 / 0.052	**+0.7 pp**	+0.2 pp
100%	0.054 / 0.057	**+0.4 pp**	+0.1 pp

Descriptive only: MAE's accuracy edge is largest at low-moderate SNR with few labels (+2.6 to +2.8 percentage points at 1–5%) and fades with more labels. Below −10 dB both are at chance (1/24 ≈ 0.042). Means over seeds; no per-SNR uncertainty.

Longer fine-tuning lifts both; the MAE edge stays inconsistent

EXPLORATORY · designed after primary test results were seen



Every run gains +0.059 to +0.086 macro-F1 over its 10-epoch run; MAE leads in 2 of 3 seeds at both lengths.

50 epochs

CNN-MAE **0.3277 ± 0.0079**

Scratch **0.3227 ± 0.0024**

Paired Δ

+0.0051 ± 0.0062

Per seed (42 / 43 / 44)

42: +0.0094

43: +0.0079

44: -0.0021

100 epochs

CNN-MAE **0.3281 ± 0.0090**

Scratch **0.3299 ± 0.0100**

Paired Δ

-0.0018 ± 0.0170

Per seed (42 / 43 / 44)

42: +0.0088

43: +0.0071

44: -0.0214

Test macro-F1, mean ± SD over 3 seeds; Δ = CNN-MAE − scratch for the same seed.

Limitations, future work and conclusion

Limitations

- No consistent MAE advantage in the primary grid
- 5% gain not significant (paired p = 0.29, 3 seeds)
- 25% and 100%: one seed each
- Longer runs: exploratory, 3 seeds, one label budget
- Noisy validation (BatchNorm statistics) affects epoch selection
- Benchmark data, not over-the-air deployment

Future work

- More paired seeds at every budget
- Stabilise validation and checkpoint selection
- Pre-specified study of mask ratio and pretraining length
- Test whether reconstruction quality predicts accuracy
- Per-SNR uncertainty estimates; controlled transfer studies

Conclusion

MAE pretraining gave modest gains in some settings: +0.012 macro-F1 at 5% labels, the low-SNR band at small budgets, and the 50-epoch follow-up. The evidence does not show a consistent advantage over training from scratch.

Take-home: with a small patch CNN, pretraining helped at the margins, but not reliably. Seeds and controls matter as much as the method.

References and questions

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[2] D. Liu, P. Wang, T. Wang, T. Abdelzaher, "Self-Contrastive Learning based Semi-Supervised Radio Modulation Classification," MILCOM 2021, pp. 777–782.

[3] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, R. Girshick, "Masked Autoencoders Are Scalable Vision Learners," IEEE/CVF CVPR, 2022, pp. 16000–16009.

[4] W. Kong et al., "A Transformer-Based Contrastive Semi-Supervised Learning Framework for Automatic Modulation Recognition," IEEE TCCN, 2023.

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[7] M. Ma, S. Liu, "Refined Semi-Supervised Modulation Classification: Integrating Consistency Regularization and Pseudo-Labeling," Future Internet, vol. 16, no. 2, art. 38, 2024.

[8] Y. Liu et al., "RIS-MAE: A Self-Supervised Modulation Classification Method Based on Raw IQ Signals and Masked Autoencoder," arXiv:2508.00274, 2025.

[9] DeepSig Inc., RadioML 2018.01A and 2016.10a datasets, https://www.deepsig.ai/datasets/.

Questions?

Live demo: Streamlit app (demo\app.py)